

Advanced Econometrics: Exercise #9

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Professor Paulo Parente

Amy Qian

Problem 1

Obtain the 95% confidence interval for the **mean** of `lhhexp1` using:

i. Standard asymptotic theory

Solution

`sum lhhexp1`

Variable	Obs	Mean	Std. dev.	Min	Max
lhhexp1	5,999	9.341561	.6877458	6.543108	12.20242

`ci means lhhexp1`

Variable	Obs	Mean	Std. err.	[95% conf. interval]	
lhhexp1	5,999	9.341561	.0088795	9.324154	9.358968

The 95% confidence interval is [9.324154, 9.358968]

ii. Standard asymptotic theory using the bootstrap mean and the bootstrap standard errors

Solution

`bootstrap r(mean), reps(499): sum lhhexp1`

First generate bootstrap sample with 499 repetitions at the mean then showing the summary.

Bootstrap results

Number of obs = 5,999
Replications = 499

Command: `summarize lhhexp1`
 `_bs_1: r(mean)`

	Observed coefficient	Bootstrap std. err.	z	P> z	Normal-based [95% conf. interval]	
_bs_1	9.341561	.0089509	1043.65	0.000	9.324018	9.359105

The 95% confidence interval is [9.324018, 9.359105].

iii. The bootstrap percentile method

Solution

`estat bootstrap, all`

This is asking STATA to show the information of all bootstrap confidence interval construction methods. The middle line is the bootstrap confidence interval using the percentile method.

$$\begin{aligned}\frac{2}{N} \sum_{i=1}^N (Y_i - \hat{b}) &= 0 \\ \frac{2}{N} \sum_{i=1}^N Y_i - \frac{2}{N} \sum_{i=1}^N \hat{b} &= 0 \\ \frac{2}{N} \sum_{i=1}^N Y_i - \frac{2}{N} N \hat{b} &= 0 \\ 2\hat{b} &= \frac{2}{N} \sum_{i=1}^N Y_i \\ \hat{b} &= \frac{1}{N} \sum_{i=1}^N Y_i\end{aligned}$$

Variable	Reps	Observed	Bias	Std. err.	[95% conf. interval]		
b_cons	499	9.341561	.0001287	.0089509	9.323975	9.359147	(N)
					9.32274	9.359476	(P)
					9.323156	9.359813	(BC)
se_cons	499	.0088795	3.52e-06	.0000952	.0086924	.0090666	(N)
					.008699	.0090717	(P)
					.0086904	.0090685	(BC)

2. regress lhhexp1

Source	SS	df	MS	Number of obs	=	5,999
Model	0	0		F(0, 5998)	=	0.00
Residual	2837.01998	5,998	.472994328	Prob > F	=	.
				R-squared	=	0.0000
				Adj R-squared	=	0.0000
Total	2837.01998	5,998	.472994328	Root MSE	=	.68775

lhhexp1	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
_cons	9.341561	.0088795	1052.04	0.000	9.324154	9.358968

3. Now use the saved bootstrap dataset to continue, in this case, bsdata.dta

The first method is the **non-symmetric test** with the critical values of $cv_{boot}(\frac{\alpha}{2})$ and $cv_{boot}(1 - \frac{\alpha}{2})$ for $(\hat{t}_1, \dots, \hat{t}_B)$.

1. gen double t_cons = (b_cons-_b[_cons])/ se_cons
2. _pctile t_cons, percentiles(2.5 97.5)
3. display _b[_cons]-_se[_cons]*r(2)
4. display _b[_cons]-_se[_cons]*r(1)

The result gives the 95% confidence interval of [9.3238664, 9.3603002].

The second method is the **symmetric test** with the critical value of $cv_{boot}^a(1 - \alpha)$ for $(|\hat{t}_1|, \dots, |\hat{t}_B|)$.

1. gen double at_cons=abs(t_cons)
2. _pctile at_cons, percentiles(95)
3. display _b[_cons]-_se[_cons]*r(1)
4. display _b[_cons]+_se[_cons]*r(1)

The result gives the 95% confidence interval of [9.3229329, 9.3601895].

Problem 2

Suppose $X_1, \dots, X_n$ are i.i.d. continuous r.v. from distribution with cumulative distribution function $F_X(\cdot)$ and probability density function $f_X(\cdot)$. Let M_n be the **sample median** and m_0 be the **population median**. In this case $\sqrt{n}(M_n - m_0) \xrightarrow{d} N\left(0, \frac{1}{4(f_X(m_0))^2}\right)$.

i. Propose a consistent estimator for the asymptotic variance of $\sqrt{n}(M_n - m_0)$

Solution

A consistent estimator for the density is $\frac{1}{4\hat{f}_x^2(M_N)}$ where $\hat{f}_x(M_N) = \frac{1}{Nh} \sum_{i=1}^N K\left(\frac{y_i - M_N}{h}\right)$, a kernel density estimator.

ii. Obtain a 95% CI for the median of lhhexp1

A. using standard asymptotic theory with the consistent estimator proposed previously

Solution

$$SE(M_N) = \frac{1}{\sqrt{n}\sqrt{4f_X^2(M_N)}}$$

$$(M_N - Z_{\frac{\alpha}{2}} SE(M_N), M_N + Z_{\frac{\alpha}{2}} SE(M_N))$$

$$P(Z > Z_{\frac{\alpha}{2}}) = \frac{\alpha}{2}, Z \sim N(0,1)$$

$h = 0.09904 \rightarrow$ obtained from Exercise sheet 8 with **Silverman's rule of thumb**

Steps to do:

1. Find the median
2. Generate the sequence
3. Find the mean of sequence

1. Do not forget to clear the dataset first as in to reload the original dataset

2. `sum lhhexp1, d`

lhhexp1				

	Percentiles	Smallest		
1%	7.586134	6.543108		
5%	8.254869	6.818342		
10%	8.507335	6.892809	Obs	5,999
25%	8.919547	6.965246	Sum of wgt.	5,999
50%	9.311416		Mean	9.341561
		Largest	Std. dev.	.6877458
75%	9.759566	11.73528		
90%	10.22297	11.75362	Variance	.4729943
95%	10.51927	12.03334	Skewness	.109243
99%	11.11577	12.20242	Kurtosis	3.599002
. * Median = 9.311416				

3. `gen double a = normalden(( lhhexp1 -9.311416)/0.09904)/0.09904`

4. `mean a`

Mean estimation			Number of obs = 5,999	
	Mean	Std. err.	[95% conf. interval]	
a	.6392072	.0153739	.6090688	.6693457

5. `display (1/(2*.6392072*sqrt(5999)))` to get $SE(M_N)$

6. `display 9.311416-1.96*0.01009925` to get lower bound of CI

7. `display 9.311416+1.96*0.01009925` to get upper bound of CI

The 95% confidence interval is [9.2916215, 9.3312105]

B. using standard asymptotic theory with the bootstrap mean and bootstrap standard error

Solution

bootstrap r(p50), reps(499): sum lhhexp1, d

Bootstrap results

Number of obs = 5,999

Replications = 499

Command: summarize lhhexp1, d
_bs_1: r(p50)

	Observed coefficient	Bootstrap std. err.	z	P> z	Normal-based [95% conf. interval]	
_bs_1	9.311416	.0113762	818.50	0.000	9.289119	9.333713

The 95% confidence interval is [9.289119, 9.333713]

C. using the bootstrap percentile method

Solution

estat bootstrap, all

Bootstrap results

Number of obs = 5,999

Replications = 499

Command: summarize lhhexp1, d
_bs_1: r(p50)

	Observed coefficient	Bias	Bootstrap std. err.	[95% conf. interval]		
_bs_1	9.3114157	-.0005968	.01137624	9.289119	9.333713	(N)
				9.286874	9.333403	(P)
				9.289034	9.334837	(BC)

Key: N: Normal
P: Percentile
BC: Bias-corrected

The 95% confidence interval is [9.2868874, 9.333403]

Problem 3

Obtain a 95% confidence interval for β_2 using

i. Standard asymptotic theory with robust standard errors

Solution

We first need to generate the squares then run the regression using robust standard error.

- gen age2= age2
- reg lnr1food lhhexp1 age age2, r

Linear regression	Number of obs	=	5,999
	F(3, 5995)	=	5484.44
	Prob > F	=	0.0000
	R-squared	=	0.8112
	Root MSE	=	.2333

lnrlfood	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
lhhexp1	.6942858	.005711	121.57	0.000	.6830901	.7054815
age	.005095	.0016213	3.14	0.002	.0019168	.0082733
age2	-.0000716	.0000161	-4.45	0.000	-.0001031	-.00004
_cons	2.1277	.0561216	37.91	0.000	2.017681	2.237718

The 95% confidence interval with robust standard error considering heteroskedasticity is [0.6830901, 0.7054815]

ii. Standard asymptotic theory using bootstrap mean and bootstrap standard errors

Solution

Now we need to use bootstrap of the pairs to answer the rest of the questions.

- `bootstrap "reg lnrlfood lhhexp1 age age2, r" _b _se, reps(499) dots saving(bsdata2)`

Bootstrap statistics	Number of obs	=	5999
	Replications	=	499

Variable	Reps	Observed	Bias	Std. err.	[95% conf. interval]		
b_lhhexp1	499	.6942858	-.0004955	.0053684	.6837384	.7048333	(N)
					.6831305	.7036634	(P)
					.683369	.7041299	(BC)
					.0019886	.0082015	(N)
b_age	499	.005095	.0000197	.0015811	.0023521	.0084537	(P)
					.0024497	.0084765	(BC)
					-.0001021	-.000041	(N)
					-.0001033	-.0000449	(P)
b_age2	499	-.0000716	-1.85e-07	.0000156	-.0001043	-.0000453	(BC)
					2.020801	2.234599	(N)
					2.029453	2.243589	(P)
					2.02276	2.238112	(BC)
b_cons	499	2.1277	.0041315	.0544089	.005337	.0060851	(N)
					.0053448	.0060896	(P)
					.0053509	.0061094	(BC)
					.0015274	.0017151	(N)
se_lhhexp1	499	.005711	-7.84e-06	.0001904	.0015204	.0017121	(P)
					.0015343	.0017242	(BC)
					.0000151	.0000171	(N)
					.000015	.0000171	(P)
se_age	499	.0016213	-3.75e-06	.0000478	.0000151	.0000171	(BC)
					.0531495	.0590936	(N)
					.0531055	.0587877	(P)
					.0533647	.0588715	(BC)

Key: N: Normal
P: Percentile
BC: Bias-corrected

We are interested in the first variable `b_lhhexp1` and its confidence intervals. The 95% confidence interval using the bootstrap mean and bootstrap standard error is [0.6837384, 0.7048333].

iii. The bootstrap percentile method based on the bootstrap of the pairs

Solution

Referring again to the previous bootstrap results. The 95% confidence interval using the bootstrap percentile method is given by [0.6831305, 0.7036634].

iv. The bootstrap t-statistics based on the bootstrap of the pairs with both methods

Solution

We now need to compute the bootstrap t-statistics using the saved database bsdata2 done in step ii.

First using the non-symmetric method:

- `reg lnrlfood lhhexp1 age age2, r`

Linear regression	Number of obs	=	5,999
	F(3, 5995)	=	5484.44
	Prob > F	=	0.0000
	R-squared	=	0.8112
	Root MSE	=	.2333

lnrlfood	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
lhhexp1	.6942858	.005711	121.57	0.000	.6830901	.7054815
age	.005095	.0016213	3.14	0.002	.0019168	.0082733
age2	-.0000716	.0000161	-4.45	0.000	-.0001031	-.00004
_cons	2.1277	.0561216	37.91	0.000	2.017681	2.237718

- `import saved database`
- `gen t_lhhexp1=( b_lhhexp1- _b[lhhexp1])/ se_lhhexp1`
- `_pctile t_lhhexp1, p(2.5, 97.5)`
- `dis _b[lhhexp1]-_se[lhhexp1]*r(2)` Lower bound
- `dis _b[lhhexp1]-_se[lhhexp1]*r(1)` Upper bound

The 95% confidence interval using the bootstrap t-statistics with non-symmetric method is given by [0.68455932, 0.70538].

Then using the symmetric method:

- `gen at_lhhexp1=abs( t_lhhexp1)`
- `_pctile at_lhhexp1, p(95)`
- `dis _b[lhhexp1]-_se[lhhexp1]*r(1)` Lower bound
- `dis _b[lhhexp1]+_se[lhhexp1]*r(1)` Upper bound

The 95% confidence interval using the bootstrap t-statistics with symmetric method (absolute value) is given by [0.68374724, 0.70482441].